

Package ‘IsoTrac’

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Title Extract intact glycoprotein masses from hRp-DIA-PTCR data using isotope matching

Version 1.0.0

Description IsoTrac deconvolutes hRp-DIA-PTCR data using isotope matching.

Its input are files prepared with PTsliCR or custom prepared data with the same formatting. It will output quality control plots, visualizations of the extracted intact masses, and detailed logs, including the intact masses.

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Encoding UTF-8

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URL <https://github.com/riley-research/IsoTrac>

BugReports <https://github.com/riley-research/IsoTrac/issues>

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magrittr,
utils,
rlang,
stats,
purrr,
enviPat,
foreach,
qpdf,
future,
future.apply,
ggplot2

LinkingTo Rcpp

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extract_isotopes

*Extract Isotope Fit Patterns from Mass Spectrometry Data***Description**

Determines intact masses from the dataset imported using the `import_spectra` function, by iteratively matching molecular isotope patterns. Processing can be executed sequentially or in parallel using a future backend. Matched and unmatched spectra profiles are written out as binned CSV tables and visual PDF spectrum plots.

Usage

```
extract_isotopes(
  full_mgf,
  aa_seq,
  protein_mass,
  threshold,
  output_folder,
  score = c("pearson", "cosine", "NRMSE"),
  number_of_cores = 1,
  isotope_peaks_included = 3,
  cutoff = 0.05,
  ppm_tolerance = 20,
  cosine_score_correction = 1,
  minimal_isotope_sequence = c(-2, -1, 0, 1, 2),
  nrmse_cutoff = 0
)
```

Arguments

<code>full_mgf</code>	<code>data.frame</code> . The data imported using the <code>import_spectra</code> function.
<code>aa_seq</code>	character. Primary amino acid sequence of the target protein backbone used for matching theoretical distribution envelopes.
<code>protein_mass</code>	numeric. Average mass of the bare protein backbone.
<code>threshold</code>	numeric. Minimum similarity fit score required to accept and subtract an isotopic envelope from the spectrum.
<code>output_folder</code>	character. Path to the parent directory where the timestamped results folder and diagnostic files will be created.
<code>score</code>	character. Vector choice containing the similarity metric to optimize. Must be one of "pearson", "cosine", or "NRMSE".
<code>number_of_cores</code>	integer. Number of parallel workers to allocate for the future multisession execution. Default is 1.
<code>isotope_peaks_included</code>	integer. Total number of isotopic peaks trailing the most intense centroided peak tracked within the calculation frame. Default is 3.
<code>cutoff</code>	numeric. General filtering score threshold limit constraint. Default is 0.05.
<code>ppm_tolerance</code>	numeric. Maximum allowable mass error assignment threshold in parts-per-million (PPM) between experimental and template centroids. Default is 20.

`cosine_score_correction`
numeric. Intensity multiplier offset constant applied during manual vector translation checks. Default is 1.

`minimal_isotope_sequence`
numeric vector. Array of required continuous isotope peaks (e.g., `c(-2, -1, 0, 1, 2)`).

`nrmse_cutoff` numeric. Normalized Root Mean Square Error limit cutoff filter. Default is 0.

Value

list. A structured list containing two data frames:

- `founddf`: Data frame containing peaks matching the template threshold bounds.
- `nonefounddf`: Data frame containing spectrum features failing template criteria.

See Also

[future::plan](#), [ggplot2::ggsave](#)

Examples

```
## Not run:
extraction_results <- extract_isotopes(
  full_mgf = experimental_data,
  aa_seq = "PEPTIDEK",
  protein_mass = 910.45,
  threshold = 0.85,
  output_folder = "./results/run_1",
  score = "cosine",
  number_of_cores = 4,
  ppm_tolerance = 15
)

## End(Not run)
```

`generate_fdr_spectra` *Generate FDR Decoy Spectra Intensities*

Description

Prepares mass spectrometry data for False Discovery Rate (FDR) estimation by manipulating peak intensities. When using the default "supervised" approach, it replaces the original centroided peak intensities with simulated values that mimic an upward trend over cycles defined by expected isotopic patterns.

Usage

```
generate_fdr_spectra(
  full_mgf,
  isotope_peaks_included,
  fdr_approach = "supervised"
)
```

Arguments

full_mgf data.frame. The data imported using the `import_spectra` function.

isotope_peaks_included integer. The number of peaks expected per isotopic cluster phase (used as the cycle length/bucket count for the simulated intensity trends).

fdr_approach character. The methodology used for decoy generation. Currently supports "supervised" to simulate bounded, cyclic trend intensities. Default is "supervised".

Value

A data.frame filtered for valid centroid entries, containing either the original or the newly simulated decoy centroided_intensity values grouped by spectra ID and slice.

Examples

```
## Not run:
# Assuming `processed_mgf` is the output from import_spectra
decoy_mgf <- generate_fdr_spectra(
  full_mgf = processed_mgf,
  isotope_peaks_included = 3,
  fdr_approach = "supervised"
)

## End(Not run)
```

get_fdr_threshold	<i>Calculate False Discovery Rate (FDR) Score Threshold</i>
-------------------	---

Description

Evaluates isotope pattern match scores for both target and decoy (FDR) spectra groups against a predicted molecular formula template. It calculates target and decoy similarity scores using single-core processing or a parallelized future backend, estimates q-values using a target-decoy approach, determines the empirical score cutoff corresponding to a target FDR, and optionally plots the score distributions.

Usage

```
get_fdr_threshold(
  full_mgf,
  full_fdr_mgf,
  aa_seq,
  protein_mass,
  score = c("pearson", "cosine", "NRMSE"),
  number_of_cores = 1,
  isotope_peaks_included = 3,
  cutoff = 0.01,
  ppm_tolerance = 20,
  cosine_score_correction = 1,
  minimal_isotope_sequence = c(-2, -1, 0, 1, 2),
  show_plot = TRUE,
  plot_bins = 0.006
)
```

Arguments

full_mgf	data.frame. The data imported using the import_spectra function.
full_fdr_mgf	data.frame. The processed decoy spectra data frame, typically generated via generate_fdr_spectra .
aa_seq	character. The primary amino acid sequence of the target protein backbone.
protein_mass	numeric. Average mass of the bare protein backbone to deduce potential glycan additions.
score	character. Similarity metric to calculate. Must be one of "cosine", "pearson", or "NRMSE" (Normalized Root Mean Square Error). Default is "pearson".
number_of_cores	integer. Number of parallel workers to allocate for processing spectra lists via <code>future.apply</code> . Default is 1.
isotope_peaks_included	integer. Number of isotropic peaks flanking the major monoisotopic peak to track inside the verification window. Default is 3.
cutoff	numeric. The False Discovery Rate constraint limit (e.g., 0.01 for a 1% FDR cutoff threshold). Default is 0.01.
ppm_tolerance	numeric. Mass accuracy window limit in parts-per-million applied when matching experimental m/z coordinates to template patterns. Default is 20.
cosine_score_correction	numeric. Intensity multiplier offset constant applied during manual cosine similarity logic computations. Default is 1.
minimal_isotope_sequence	numeric vector. Continuous isotope index steps relative to the base profile required to validate and trim isotope fit profiles. Default is c(-2, -1, 0, 1, 2).
show_plot	logical. If TRUE, prints a line plot comparing target and decoy score binned counts alongside the derived threshold boundary. Default is TRUE.
plot_bins	numeric. Bin width size used to cluster matching score counts inside the target-decoy visual plot. Default is 0.006.

Value

numeric. The calculated score threshold value corresponding to the exact target FDR allocation parameter restriction.

See Also

[generate_fdr_spectra](#), [import_spectra](#)

Examples

```
## Not run:
# Standard wrapper calculation at 1% FDR using 4 parallel processing cores:
fdr_cutoff <- get_fdr_threshold(
  full_mgf = target_spectra,
  full_fdr_mgf = decoy_spectra,
  aa_seq = "PEPTIDEK",
  protein_mass = 910.45,
  score = "pearson",
  number_of_cores = 4,
```

```

    cutoff = 0.01
)

## End(Not run)

```

import_spectra

Import and Process Mass Spectrometry Spectra

Description

Reads a comma-separated text file (representing MGF spectral data), as generated by PTsliCR, with the columns m/z, intensity, id, peakgroup. The importer applies optional scan range filtering, filters out low-intensity signals, segments data into distinct mass-to-charge (m/z) separation slices, performs profile-to-centroid peak detection using spline interpolation, determines charge states based on delta m/z differences, and estimates the intact mass.

Usage

```

import_spectra(
  mgf_path,
  intensity_threshold,
  charge_range,
  separation_threshold = 20,
  min_peaks_per_slice = 6,
  min_charge_percentage_matched = 0.2,
  scan_range = c(0, 0)
)

```

Arguments

mgf_path	character. Path to the comma-separated text file containing the spectrum data.
intensity_threshold	numeric. Minimum intensity threshold. Peaks falling completely below this threshold are filtered out.
charge_range	integer vector. A vector of valid expected charge states to evaluate (e.g., 1:5).
separation_threshold	numeric. The minimal difference in m/z required to split spectral peaks into a new sequential slice. Default is 20.
min_peaks_per_slice	integer. Minimum number of peaks required within a slice to keep it. Default is 6.
min_charge_percentage_matched	numeric. The minimum proportion of sequential peak differences that must match a charge state's calculated delta for the assignment to be retained. Default is 0.2.
scan_range	numeric vector. A vector of length 2 indicating the minimum and maximum spectrum_nr to keep. Use c(0, 0) to skip filtering. Default is c(0, 0).

Value

A `data.frame` (or `tibble`) containing the processed spectrum data, including calculated centroids, assigned charge states, peak counts, slice segmentation, and the estimated intact mass (`estimated_intact_mass`).

Examples

```
## Not run:
# Example workflow usage:
processed_data <- import_spectra(
  mgf_path = "path/to/data.csv",
  intensity_threshold = 1000,
  charge_range = 1:4,
  separation_threshold = 20,
  min_peaks_per_slice = 5,
  min_charge_percentage_matched = 0.25,
  scan_range = c(100, 500)
)

## End(Not run)
```

`plot_spectra`*Plot Mass Spectrometry Spectra to PDF*

Description

This function iterates through unique spectrum IDs inside the mass spec data, and exports a single multi-page PDF document containing line plots for each spectrum. The number above each PTCR slice are the determined charge states for that slice.

Usage

```
plot_spectra(full_mgf, output_folder)
```

Arguments

<code>full_mgf</code>	<code>data.frame</code> . The data imported using the import_spectra function.
<code>output_folder</code>	A character string specifying the directory path where the output PDF "all_spectra_plot.pdf" should be saved.

Value

No return value. The function is called for its side effect of writing a PDF file to the specified `output_folder`.

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